



Mini Project (BCS-752)

**Mini Project Presentation
On
Business KPI Assistant**

**Submitted by
Muskan Kumari – 2300971640051
Vanshika Barua – 2300971640069
Anshul Kushwaha – 2300971640018
Harshit Chauhan – 2300971640038**

**Under the Supervision
of
Mr. Ajeet Kumar**

**Galgotias College of Engineering & Technology
Greater Noida, Uttar Pradesh**



Table of Content

- Introduction
- Literature survey
- Problem statement
- Objective
- Conclusion



Introduction

- Business organizations work with structured datasets containing information such as products, sales, revenue, profit, quantity, regions, customer segments, and dates.
- Traditional business analysis often requires manual data cleaning, calculations, visualization, and interpretation.
- This process can be time-consuming and difficult for non-technical users.
- **Business KPI Assistant** is proposed as an AI-powered analytics tool for simplified business data analysis.
- Users can upload business datasets and interact with them using **natural-language questions**.
- The system analyzes queries, identifies relevant metrics and data fields, and performs the required analysis.
- It provides **KPI calculations, statistical analysis, trend identification, anomaly detection, comparative analysis, and visualizations**.
- Analytical results are presented through **understandable natural-language business insights**.



Literature survey

Natural Language-Based Data Analytics

- Enables users to interact with structured business data using conversational questions.
- Identifies user intent, relevant data fields, metrics, aggregation, and analytical operations.

Automated KPI Identification

- Automatically calculates important business metrics such as:
 - Revenue and Sales
 - Profit and Profit Margin
 - Average Order Value
 - Units Sold
 - Revenue and Sales Growth

AI-Based Query Understanding

- Converts natural-language questions into structured analytical operations.
- Supports filtering, grouping, aggregation, sorting, comparison, statistical analysis, and time-series analysis.



Literature survey

Automated Visualization

- Selects suitable visualizations according to the analytical question.
- Supports charts such as bar, line, pie/donut, area, scatter, KPI cards, and heatmaps.

Trend & Anomaly Analysis

- Identifies sales and revenue trends, growth patterns, seasonal patterns, and unusual observations.

Explainable & Conversational Insights

- Converts analytical findings into understandable explanations.
- Supports follow-up questions on the same dataset for deeper exploration.



Problem statement

Traditional business data analysis requires users to manually:

- Clean and prepare datasets
- Write formulas and perform calculations
- Create appropriate charts
- Analyze and interpret numerical results
- Prepare reports repeatedly

This creates challenges for **non-technical users**, especially when working with large Excel or CSV datasets.

Business users may need answers to questions such as:

- Which product generated the highest sales?
- Which region generated the most revenue?
- Why did revenue decrease during a particular period?
- What is the monthly sales trend?
- Are there any unusual sales values?

Therefore, the **Business KPI Assistant** is proposed to simplify business analysis through natural-language interaction and automated analytical processing.



Objectives

- Enable users to upload business datasets in **CSV and Excel formats**
- Process datasets by examining structure, columns, data types, and missing values.
- Allow users to ask **natural-language questions** about business data.
- Understand user intent and identify relevant metrics, dimensions, and data fields.
- Automatically calculate important **business KPIs**.
- Perform filtering, grouping, aggregation, sorting, statistical, comparative, and time-series analysis.
- Generate suitable visualizations based on the analytical question.
- Identify business **trends, patterns, and seasonal changes**.
- Detect unusual observations and business anomalies.
- Perform comparative analysis across products, regions, months, quarters, years, and target values.
- Generate understandable **natural-language business insights**.
- Provide an interactive analytics dashboard for exploring business information.
- Support **conversational follow-up questions** on the same dataset.



Conclusion

- **Business KPI Assistant** provides an AI-powered approach to analyzing structured business datasets through natural-language interaction.
- The proposed system simplifies the traditional process of data cleaning, KPI calculation, analysis, visualization, and interpretation.
- It enables users to obtain meaningful business information without manually writing formulas or analytical queries.
- The system integrates **KPI analysis, trend identification, anomaly detection, comparative analysis, visualization, and natural-language insights**.
- Conversational follow-up questions allow users to continue exploring the same dataset and perform deeper analysis.
- Overall, the proposed system aims to simplify business data analysis and support **data-driven decision-making**.



Thank You